Sudoku solver 1

Saturday, January 3, 2026 07:13 PM

3 conditions ⇒

1. Row ⇒ All values 1–9 are present & unique
2. Col ⇒ All values 1–9 are present & unique
3. 3x3 ⇒ In 3x3 matrix all 1–9 ele are present & unique.

	0	1	2	3	4	5	6	4	8
0	4	5	1						
1			2		7		6	3	
2								2	8
3				9	5				
4		8	6				2		
5		2		6			7	5	
6							4	7	6
7		7			4	5			
8			8			9			

Code ⇒ (no base case)

```
solve ( 2-D arr sudoku , int n)
{
    for ( i = 0 ⟶ n)
    {
        for ( j = 0 ⟶ n)
        {
            if ( sudoku[i][j] == 0)
            {
                for ( value = 1 ⟶ 9)
                {
                    if ( isSafe ( sudoku, n, i, j, value))
                    {
                        sudoku[i][j] = value;
                        bool aagekaSol = solve(sudoku, n);
                        if (aagekaSol == true)
                        {
                            return true;
                        }
                        // backtrack
                        sudoku[i][j] = 0;
                    }
                }
```

```
}
return false;
}
}
}
return true;
}
```

Sudoku solver 2

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	0	1	2	3	4	5	6	4	8
0	4	5	1						
1			2		7		6	3	
2								2	8
3				9	5				
4		8	6				2		
5		2		6			7	5	
6							4	7	6
7		7			4	5			
8			8			9			

isSafe(sudoku, n, row, col)

→ Condi → row → Mujhe saari row nahi check karni

i = 0 row

i = 0, j = 0

```
while (j < n)
{ if (sudoku[i][j] == value)
  { return false;
  }
  j++;
}
```

⇒ Con2 ⇒ Col | i = 0, col = 0 + col

```
while (i < n) { if (sudoku[i][j] == value)
                { return false;
                }
                i++;
}
```

⇒ Cond^n 3 ⇒ (3x3 unique value)

```
for (int i=0; i<9; i++)
{ if (board[3*(row/3) + (i/3)][3*(col/3) + (i%3)] ==
        value)
  { return false;
  }
}
return true;
}
```

Sudoku solver 3

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	0	1	2	3	4	5	6	7	8
0	4	5	1						
1			2		7		6	3	
2								2	8
3				9	5				
4		8	6				2		
5		2		6			7	5	
6							4	7	6
7		7			4	5			
8			8			9			

3rd Condn

⇒ Sudoku[3*(row/3)+(i/3)][3*(col/3)+(i%3)]

⇒ Sudoku[4][5]

⇒ row = 4
col = 5

let i = 0 ⇒
i = 1

$3\times(\frac{4}{3})+(\frac{0}{3})$, $3\times(\frac{5}{3})+(0\%3)$

$3\times1+0$, $3\times1+0$

(3 , 3 → (3,3)

$3\times(\frac{4}{3})+(\frac{1}{3})$, $3\times(\frac{5}{3})+(1\%3)$

3 + 0 , 3 + 1

(3, 4)

i = 2 (3, 5)

i = 3 $3\times(\frac{4}{3})+(\frac{3}{3})$, $3\times(\frac{5}{3})+(3\%3)$

3 + 1 , 3 + 0

(4, 3)

Sudoku solver 4

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isSafe logic in one loop → 0 → 9

```
for ( i = 0 ———————→ n or 9)
{
    row // =>
    if ( sudoku[row][i] == value) { return false; }
    // col =>
    if ( sudoku[i][col] == value) { return false; }

    3x3
    if ( sudoku[3x(row/3)+(i/3)][3x(col/3)+(i%3)] == value
        { return false;
        }
}
return true.
```